Aishwarya Aravind Nair

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Education

University of Wisconsin - Madison — MS Information | GPA: 3.8/4.0 Sept 2023 – May 2025

University of Mumbai — BE Instrumentation Engineering Aug 2017 – June 2021

Skills

Certifications: Azure Administrator Associate (AZ-104)

Azure: VMs, AKS, App Gateway, Entra ID, Azure DevOps, Azure Automation, Azure Cost Management

Identity & Access: Azure Entra ID (SSO, Conditional Access, Identity Lifecycle), Active Directory, LDAP, SAML, OAuth 2.0

Cloud & Infra as Code: AWS (EKS, Lambda, EC2, S3, DynamoDB, CloudWatch), Terraform, PowerShell, Azure CDK

Containers & OS: Docker, Kubernetes, Helm, Linux (RHEL, Ubuntu, Debian), Windows Server

Monitoring & Governance: Prometheus, Grafana, ELK Stack, CloudWatch, incident response, cost optimization

DevOps & CI/CD: GitHub Actions, Azure DevOps, GitLab, Helm-based pipelines

Languages: Python, PowerShell, JavaScript, SQL

Experience

DevOps Engineer Intern, XNODE Inc. — Remote Oct 2025 – Present

- Enhanced Model Context Protocol (MCP) server to support HTTP/REST API endpoints (modified from stdio-only architecture) using Claude Code for debugging and refactoring; containerized on EKS and deployed via Azure DevOps CI/CD to enable client AI workflow integration
- Optimized Azure DevOps CI/CD pipeline for EKS deployments by implementing Docker layer caching, multi-stage builds, and job parallelization; reduced deployment time from 15 to 6 minutes
- Managed infrastructure provisioning using Terraform across dev/staging/prod environments; ensured consistent infrastructure configuration and maintained infrastructure-as-code best practices
- Deployed and managed HashiCorp Vault for credential management; tested application deployments and documented operational procedures for production deployments
- Contributed to Prometheus and Grafana monitoring setup; tracked EKS cluster health and resource utilization

Cloud Engineer, Assistant System Engineer, Tata Consultancy Services — Mumbai Jun 2021 – Aug 2023

- Migrated manual deployment process to automated CI/CD pipeline using Terraform, Python, and Azure DevOps for fintech platform processing 50M+ monthly transactions, reducing release windows from 5 hours to 2 hours; automated deployment process using Terraform modules, standardizing infrastructure provisioning across dev/staging/prod environments
- Architected and deployed Azure App Gateway with AKS cluster supporting 99.99% uptime during peak loads; implemented SSL/TLS termination, rate limiting, health monitoring, and horizontal pod autoscaling
- Executed MySQL migrations from Single Server to Flexible Server using hybrid approach: online DMS replication for UAT environment; offline approach (full backup + incremental sync during cutover) for production due to large database size and replication lag concerns; achieved less than 5 minutes cutover time with zero data loss
- Engineered automated log archival solution for financial transaction audit trail using Python and Azure Blob Storage (Cool Tier); moved 100GB/month of application logs while maintaining 7-year retention compliance, reducing storage costs by 40%
- Built Azure Monitor dashboards and automated VM scheduling policies for non-production environments, reducing monthly cloud spend by 15% and improving operational visibility across 50+ resources

Projects

Job Alert Agent | Python, FAISS, Groq LLM, Apify, Google Sheets API, Streamlit, GitHub Actions 2025

- Developed job matching system combining semantic search (FAISS) and LLM-powered role classification using Groq LLaMA-3.3; initially built custom scraper but switched to Apify for reliability across LinkedIn and Indeed
- Engineered data pipeline with Google Sheets API for persistent storage and built Streamlit frontend for job browsing; automated end-to-end workflow via GitHub Actions for daily scraping, matching, and categorization

Task Management REST API | Python/FastAPI, PostgreSQL, Redis, Docker 2025

- Built RESTful API with JWT authentication, role-based access control, and request validation; containerized with Docker and documented with Swagger UI
- Designed normalized database schema with proper indexing; implemented Redis caching for frequently accessed resources and wrote unit/integration tests using pytest

Event-Driven Processing Pipeline | Kafka, Python, PostgreSQL, Docker 2025

- Built event streaming pipeline using Kafka with consumer groups for parallel processing; implemented dead letter queue for handling failed messages
- Designed fault-tolerant consumer logic with proper offset management ensuring no message loss; added Grafana dashboards tracking consumer lag, message throughput, and error rates